The code includes four configurable Boolean flags that control the execution of different stages of the processing pipeline:

- `dodet1 = True` → Executes calwebb_detector1
- `doimage2 = True` → Executes calwebb_image2
- `doimage3 = True` → Executes calwebb_image3
- `remove_1group_uncal = True` → Removes the `uncal_1group` directory at the end of execution to reduce disk usage

In addition, the parameter
`unlimit_desat = 0`
controls the desaturation strategy:

- `0`: No desaturation modification (default pipeline behavior).
- `1`: Apply a 1-group desaturation only.
- `2`: Execute the full pipeline and additionally perform the 1-group desaturation.

The desaturation procedure is governed exclusively by `unlimit_desat`. The `remove_1group_uncal` flag only affects storage management and does not alter the scientific processing.

Pipeline Version and Processing Stages

The implementation corresponds to the updated 2025 version of the JWST Calibration Pipeline, executing Stages 1, 2, and 3 under standard configurations. Users are advised to verify the individual processing steps within each stage to ensure consistency with current calibration standards.

Desaturation Methodology

The only modification relative to the standard pipeline is the option to control the number of detector groups used during processing:

- **Standard mode (`unlimit_desat = 0`)**: All available groups are used.
- **Low-exposure mode (`unlimit_desat = 1 or 2`)**: The number of groups is reduced to one, effectively simulating the shortest possible exposure time to mitigate saturation effects.

This approach can recover information in highly saturated regions, such as auroral regions or the Great Red Spot on Jupiter.

The method was particularly useful for analyses such as those performed by Ricardo Hueso, whose cloud-tracking algorithm relies on relative intensity contrasts rather than absolute radiometric calibration. However, when operating in 1-group mode, the absolute radiance values may not be reliable, and results should therefore be interpreted with caution if absolute fluxes are required.

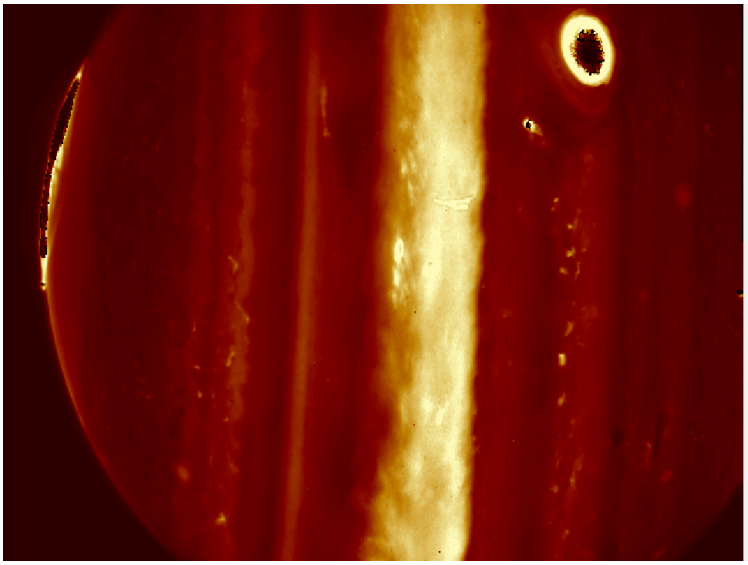
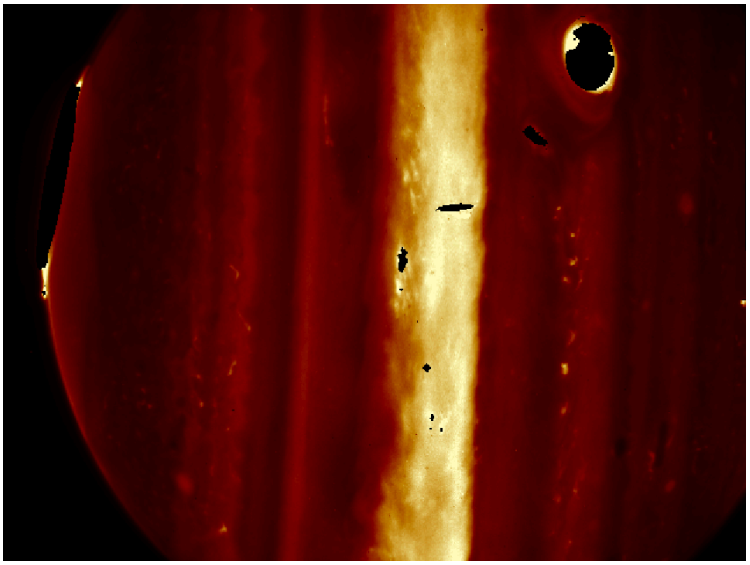
Technical Implementation

When `unlimit_desat = 1` or `2`, the code modifies the `uncal.fits` files by:

1. Reducing the number of groups to one (or to the number defined by the variable `k` in line 132 of the script).
2. Updating the FITS header keyword `NGROUPS` to ensure compatibility with internal pipeline requirements.
3. Replacing the original multi-group file (e.g., 5 groups) with the modified single-group version prior to further processing.

A comparison analysis is provided to illustrate how this strategy enhances signal recovery in strongly saturated regions while preserving the capability to run the full calibration sequence.

1 group



4 groups